

# How Do Socio-Economic Factors Influence Global Life Expectancy?

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## Introduction

Global life expectancy is widely used as a key indicator of population health and overall quality of life. Despite steady global improvements over the past several decades, substantial disparities persist between countries. These differences are often driven by socio-economic factors such as income level, education investment, and healthcare spending.

### Research Question:

*How have life expectancy trends evolved globally over the last two decades, and which socio-economic factors are most strongly associated with these trends?*

The purpose of this analysis is to evaluate how selected socio-economic factors are associated with life expectancy across countries between 2000 and 2019. In particular, this report examines whether differences in income classification and national investments in health and education are statistically associated with differences in life expectancy. The results of this analysis are intended to support high-level policy and resource-allocation decisions.

## Dataset

The dataset used in this analysis comes from Kaggle – “Life Expectancy & Socio-Economic Factors” (World Bank) (<https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank>), which contains indicators collected by the World Bank. The dataset includes observations from multiple countries between 2000 and 2019.

### Data Cleaning

In order to prepare the data for analysis, we executed the following data cleaning steps:

- Standardized column names
- Removed duplicates and empty strings
- Convert numeric-looking strings to numeric
- Filtered years 2000–2019
- Dropped rows with missing `Life.expectancy`
- Saved cleaned dataset as `cleaned_life_expectancy.csv`

```
## File saved: cleaned_life_expectancy.csv
```

### Key Variables

The analysis focuses on the following key variables:

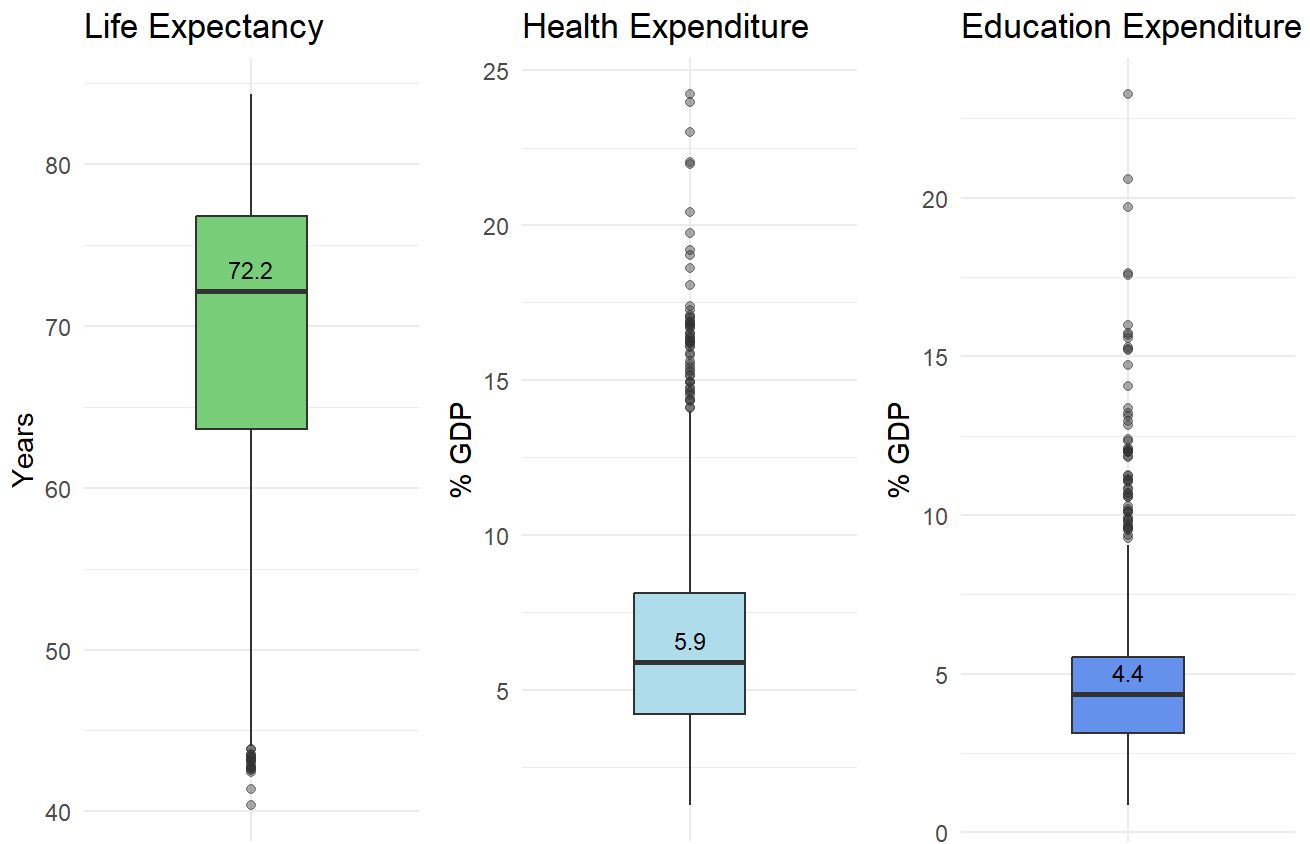
| Variable                          | Description                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Life.Expectancy.World.Bank</b> | Average life expectancy at birth, reported by the World Bank.                                              |
| <b>Health.Expenditure..</b>       | Current health expenditure as a percentage of GDP, representing spending on healthcare services and goods. |
| <b>Education.Expenditure..</b>    | Total government expenditure on education as a percentage of GDP, including current and capital spending.  |
| <b>IncomeGroup</b>                | World Bank income classification (Low, Lower Middle, Upper Middle, High).                                  |

## Descriptive Statistics

In order to understand the spread of the data, we looked at descriptive statistics for the selected columns:

```
## Life.Expectancy.World.Bank Health.Expenditure.. Education.Expenditure..
## Min. :40.37 Min. : 1.264 Min. : 0.8503
## 1st Qu.:63.64 1st Qu.: 4.205 1st Qu.: 3.1361
## Median :72.17 Median : 5.892 Median : 4.3715
## Mean :69.75 Mean : 6.364 Mean : 4.5890
## 3rd Qu.:76.81 3rd Qu.: 8.119 3rd Qu.: 5.5198
## Max. :84.36 Max. :24.231 Max. :23.2700
## NA's :188 NA's :180 NA's :1090
## IncomeGroup
## Length:3306
## Class :character
## Mode :character
##
##
##
##
```

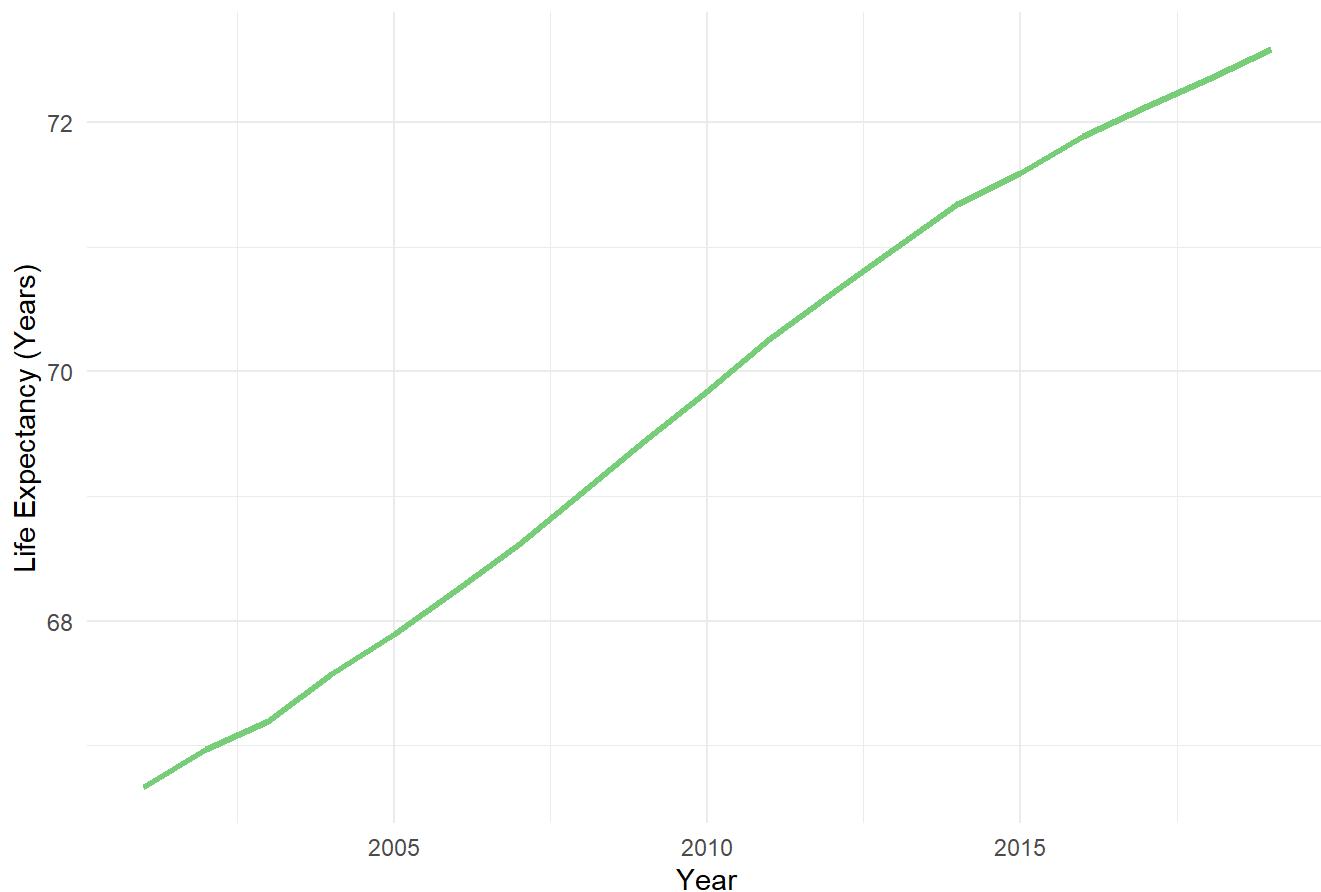
This is what it looks like visually:



The statistics reveal that life expectancy, health expenditure, and education expenditure vary considerably across countries, reflecting global inequalities. Health expenditure shows slightly higher variability relative to the median than education spending. However, missing data is highest for education expenditure, so analyses involving this variable may have reduced sample size.

Then, we wanted to understand what the spread of life expectancy looks like over time:

## Average Global Average Life Expectancy (2000–2019)



Life expectancy has generally increased worldwide over the last two decades. While most countries show improvements, some countries lag behind, showing that gains are not the same everywhere.

## Hypotheses

Based on the observed trends in the descriptive statistics, we formulated some research questions to assess how socio-economic variables contributed to the increase in life expectancy:

- How Does Health Expenditure Influence Life Expectancy?
- How Does Education Expenditure Influence Life Expectancy?
- How Does Life Expectancy Differ By Income Group?

We wanted to dive deeper and explore how strong are the relationships between these variables. Hence, we developed the following hypotheses:

### Hypothesis 1: Health Expenditure vs. Life Expectancy

- **Null Hypothesis ( $H_0$ ):** Health expenditure is *not* associated with life expectancy.
- **Alternative Hypothesis ( $H_1$ ):** Higher health expenditure *is* associated with higher life expectancy.

### Hypothesis 2: Education Expenditure vs. Life Expectancy

- **Null Hypothesis ( $H_0$ ):** Education expenditure is *not* associated with life expectancy.
- **Alternative Hypothesis ( $H_1$ ):** Higher education expenditure *is* associated with higher life expectancy.

### Hypothesis 3: Life Expectancy by Income Group

- **Null Hypothesis ( $H_0$ ):** Mean life expectancy is *equal* across all income groups.
- **Alternative Hypothesis ( $H_1$ ):** At least one income group has a *different* mean life expectancy.

## Statistical Analysis

To evaluate the proposed hypotheses, we selected appropriate statistical methods based on the data types and the research questions.

To examine the relationships between life expectancy and continuous socio-economic indicators, we decided to apply simple linear regression. Life expectancy served as the outcome variable, while health expenditure and education expenditure were analyzed separately as predictor variables. Linear regression was chosen because it allows us to estimate of both the direction and strength of association between a continuous predictor and a continuous outcome.

To assess whether average life expectancy differs across economic development levels, a one-way analysis of variance (ANOVA) was conducted using World Bank income groups as the categorical predictor. ANOVA is appropriate when comparing mean outcomes across more than two groups and allows for testing whether at least one group mean differs significantly from the others.

## Test Results

### 1. Health Expenditure Regression

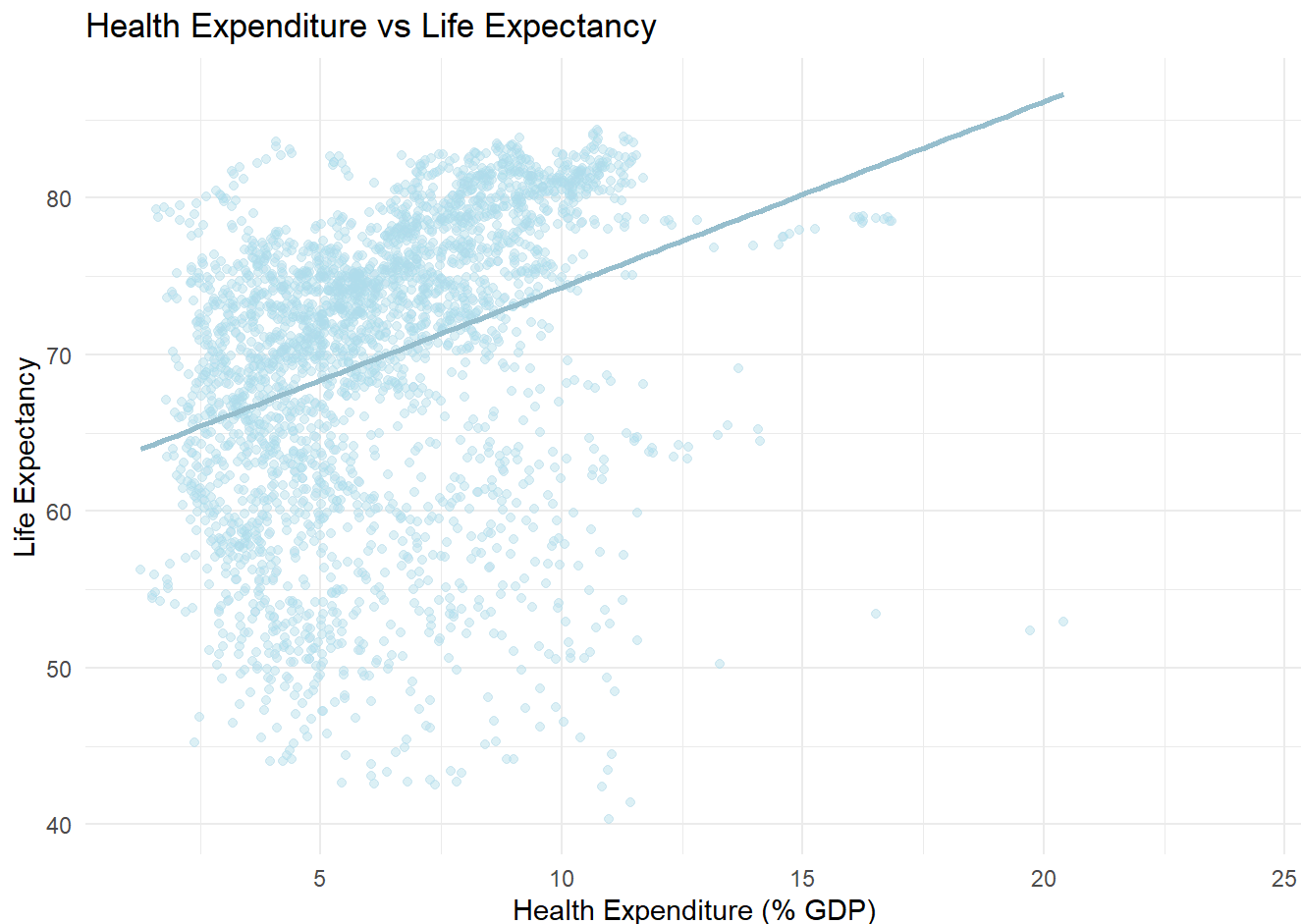
```
##
## Call:
## lm(formula = Life.Expectancy.World.Bank ~ Health.Expenditure..,
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.082  -4.474   2.678   6.434  16.305
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    62.45581    0.42255  147.81  <2e-16 ***
## Health.Expenditure.. 1.18436    0.06289   18.83  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.766 on 2974 degrees of freedom
## (330 observations deleted due to missingness)
## Multiple R-squared:  0.1065, Adjusted R-squared:  0.1062
## F-statistic: 354.6 on 1 and 2974 DF, p-value: < 2.2e-16
```

A simple linear regression was conducted to examine the relationship between national health expenditure and life expectancy. The results indicate that health expenditure is a statistically significant predictor of life expectancy,  $F(1, 2974) = 354.6$ ,  $p < .001$ .

The regression coefficient for health expenditure was positive and statistically significant ( $\beta = 1.18$ ,  $p < .001$ ), indicating that, on average, a one percentage point increase in health expenditure (as a percentage of GDP) is associated with an increase of approximately 1.18 years in life expectancy. The model explained approximately 10.6% of the variance in life expectancy ( $R^2 = 0.106$ ).

Based on these results, *the null hypothesis was rejected*. There is strong statistical evidence to support the conclusion that higher health expenditure is associated with higher life expectancy.

This is what it looks like visually:



Countries that spend more on healthcare tend to have higher life expectancy.

## 2. Education Expenditure Regression

```
##
## Call:
## lm(formula = Life.Expectancy.World.Bank ~ Education.Expenditure..,
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.380  -5.350   2.354   6.844  15.343
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      65.7326     0.5062 129.848  <2e-16 ***
## Education.Expenditure..  1.0187     0.1035   9.845  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.95 on 2143 degrees of freedom
## (1161 observations deleted due to missingness)
## Multiple R-squared:  0.04327,    Adjusted R-squared:  0.04283
## F-statistic: 96.93 on 1 and 2143 DF,  p-value: < 2.2e-16
```

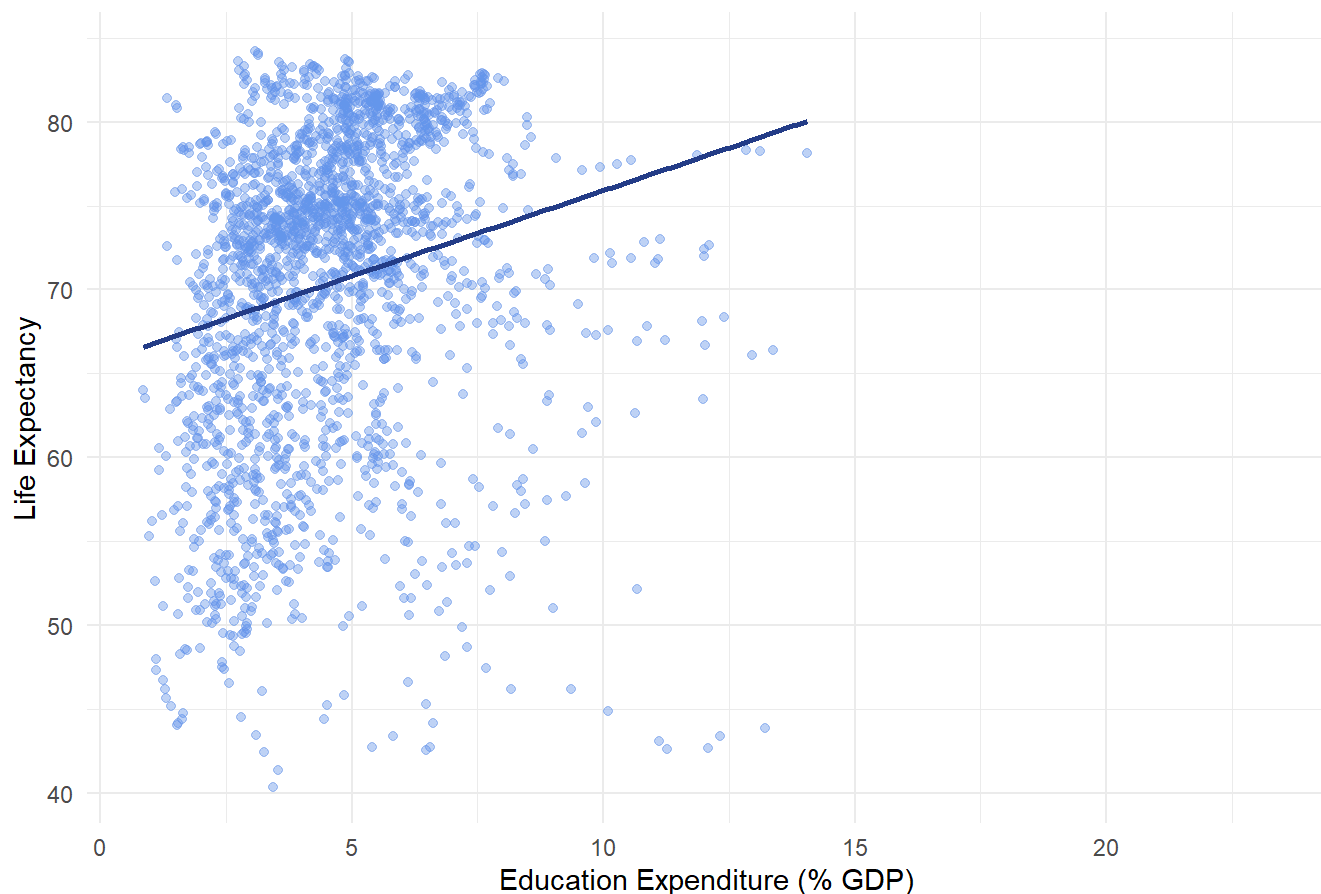
A simple linear regression was also conducted to assess the relationship between education expenditure and life expectancy. The analysis revealed a statistically significant positive association between education expenditure and life expectancy,  $F(1, 2143) = 96.93$ ,  $p < .001$ .

The estimated regression coefficient for education expenditure was positive ( $\beta = 1.02$ ,  $p < .001$ ), suggesting that a one percentage point increase in education expenditure is associated with an increase of approximately 1.02 years in life expectancy. However, the model explained a smaller proportion of the variance ( $R^2 = 0.043$ ), meaning that education expenditure alone accounts for about 4.3% of the variation in life expectancy.

As a result, *the null hypothesis was rejected*. While education expenditure is significantly associated with life expectancy, the strength of this relationship is weaker compared to health expenditure.

This is what it looks like visually:

### Education Expenditure vs Life Expectancy



Higher education spending is generally associated with longer life expectancy, but the relationship is weaker and less consistent than with health spending.

## 3. Income Group and Life Expectancy (ANOVA)

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## IncomeGroup   3 169721   56574   1659 <2e-16 ***
## Residuals 3114 106175     34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 188 observations deleted due to missingness
```

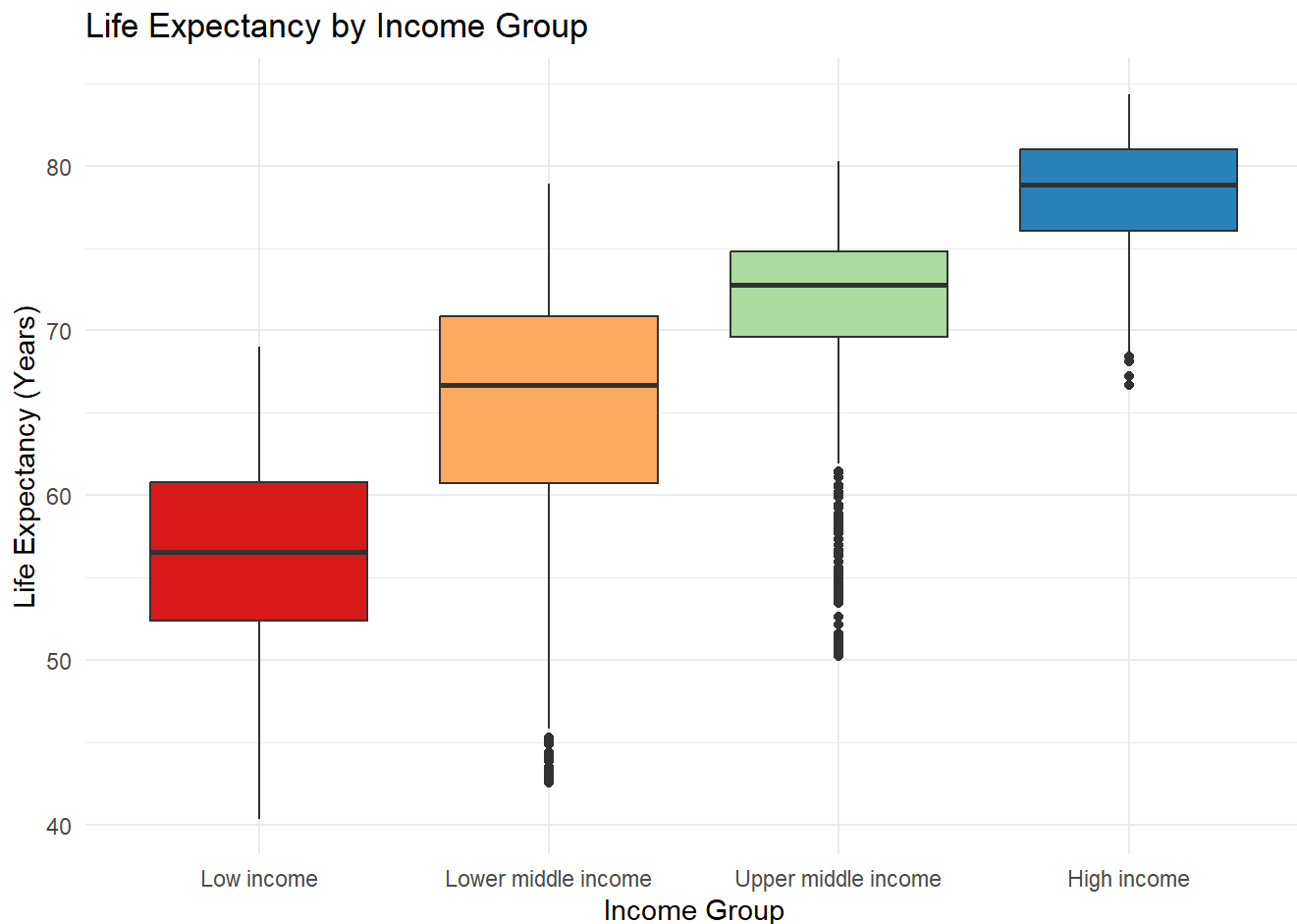
A one-way analysis of variance (ANOVA) was performed to compare mean life expectancy across World Bank income groups (low income, lower middle income, upper middle income, and high income). The results show a statistically significant difference in mean life expectancy between income groups,  $F(3, 3114) = 1659$ ,  $p < .001$ .

These findings indicate that at least one income group differs significantly from the others in terms of average life expectancy. Examination of group distributions shows a clear increasing trend in life expectancy as income classification rises.

Accordingly, the null hypothesis was rejected. Income group is strongly associated with differences in life expectancy across countries.

This is what it looks like visually:





Life expectancy clearly increases as income level rises. High-income countries have the highest life expectancy, and low-income countries have the lowest, matching the statistical results.

## Insights

### Interpretation of Results

The analysis shows clear positive relationships between socio-economic conditions and global life expectancy.

Health expenditure has the strongest and most consistent relationship, highlighting that better-funded healthcare systems (such as hospitals, medical staff, and preventive care) support longer lifespans. Education expenditure also contributes to higher life expectancy, though the effect is smaller and more variable, suggesting indirect pathways such as improved health literacy and economic opportunities.

Life expectancy differs significantly across World Bank income groups, with higher-income countries generally living longer. This reflects differences in healthcare access, living conditions, and social infrastructure.

These insights reveal that while national wealth explains broad differences between countries, targeted investments in healthcare and education play a meaningful role in shaping health outcomes within and across income groups.

## Decision-Making Implications

Policymakers can use these findings to prioritize interventions. For lower-income and middle-income countries, increasing health expenditure may yield immediate improvements in life expectancy, particularly through primary care, disease prevention, and maternal and child health services. Education investment supports long-term health improvements by fostering better-informed and economically empowered populations. Coordinated strategies balancing both healthcare and education spending are likely to be most effective in reducing global health disparities.

## Data Limitations

This analysis is subject to several limitations:

First, the dataset is aggregated at the national level and does not capture individual-level health outcomes or inequalities within countries. As a result, regional and demographic variations in life expectancy are not reflected in the analysis.

Second, some potentially relevant socio-economic indicators, such as income inequality, healthcare access metrics, and environmental factors, were excluded to maintain analytical simplicity.

Additionally, missing data across years and countries reduced the sample size, which may introduce bias and limit generalizability.

Finally, it is important to note that this is merely a statistical observation of relationships, so the identified associations should not be interpreted as absolute direct causes.

## Conclusion

Overall, this analysis confirms that socio-economic factors strongly influence global life expectancy. Higher-income countries tend to have longer lifespans, and targeted investments in health and education further enhance population health. As a result, policymakers aiming to improve life expectancy should prioritize healthcare funding while maintaining long-term investments in education. Together, national wealth and strategic socio-economic investments play crucial roles in promoting longer, healthier lives worldwide.